

Independent replication of Rötteler & Beth (1998):
Polynomial-Time Solution to the Hidden Subgroup Problem
for a Class of non-Abelian Groups
(arXiv:quant-ph/9812070)

QC-200 replication wave

2026-07-05

Abstract

We independently reproduce the core quantum algorithm of Rötteler and Beth (quant-ph/9812070) for the hidden subgroup problem (HSP) on the wreath products $W_n := \mathbb{Z}_2^n \wr \mathbb{Z}_2$ of order 2^{2n+1} . Working entirely in numpy on the group algebra of W_n for $n = 2$ ($|G| = 32$) and $n = 3$ ($|G| = 128$), we implement (a) the group multiplication, (b) the non-abelian pairing μ (Def. 4.3), (c) the Fourier matrix DFT_{W_n} with entries $(-1)^{\mu(g,h)}/\sqrt{|G|}$, (d) coset-state preparation, and (e) the two-stage sample-and-nullspace postprocessing from Algorithm 7.1. Across 6 initial random subgroups (three with $|H| = 2$, three with $|H| = 4$) at $n = 2$, the algorithm recovers H with empirical success probability 1.00 using 32 samples per run, well above the paper’s $4n = 8$ expected query bound. A stress sweep of samples from 2 to 32 tracks the paper’s Lemma 6.3 bound $P(\text{success}) \geq 1 - 2^{-i/4}$, with the empirical 0.9-crossover at 8–12 samples matching the theoretical prediction. Non-abelian-focused trials with transversal ($a=1$) generators, which require the wreath-specific Fourier machinery rather than the abelian Simon subroutine, likewise succeed at 1.00. A scaling test at $n = 3$ recovers H with 1.00 probability using $4n = 12$ samples, confirming the paper’s polynomial-in- n query claim. **Verdict: REPLICATED.**

1 Paper summary

Rötteler and Beth exhibit a family of non-abelian groups on which the hidden subgroup problem admits an exact quantum polynomial-time algorithm. The family is the wreath product

$$W_n := \mathbb{Z}_2^n \wr \mathbb{Z}_2,$$

whose elements are triples $(x, y; a)$ with $x, y \in \mathbb{F}_2^n$ and $a \in \mathbb{F}_2$. Multiplication is (paper p. 4):

$$(x, y; a) \cdot (x', y'; a') = \begin{cases} (x \oplus x', y \oplus y'; a \oplus a') & a' = 0, \\ (y \oplus x', x \oplus y'; a \oplus a') & a' = 1. \end{cases}$$

$|W_n| = 2^{2n+1}$. The base group is $N = \mathbb{Z}_2^n \times \mathbb{Z}_2^n$ (index 2 normal subgroup), and every subgroup $U \leq W_n$ factors as $U = (U \cap N) \cdot (U \cap U^t)$ where $t = (0, 0; 1)$ (Lemma 4.1). The paper defines a symmetric pairing $\mu : W_n \times W_n \rightarrow \mathbb{F}_2$ (Def. 4.3) that plays the role of the character pairing in the abelian case, and shows that the group Fourier transform

$$DFT_{W_n} = \frac{1}{\sqrt{|W_n|}} [(-1)^{\mu(g,h)}]_{g,h \in W_n}$$

maps a uniform superposition over any coset g_0U into a uniform superposition (with ± 1 phases) over $U^\perp$ if $g_0 \in N$ and over $(U^t)^\perp$ if $g_0 \notin N$ (Thm. 6.2 and following remark).

Algorithm 7.1 then proceeds by (i) running the abelian HSP algorithm on N to recover $U \cap N$, and (ii) iterating quantum coset-state preparation and Fourier sampling to collect $O(n)$ elements of $\langle U^\perp, (U^t)^\perp \rangle$, whose orthogonal complement under μ is $U \cap U^t$ (Cor. 4.7). Combining the two gives U via the Lemma 4.1 factorization. Lemma 6.3 bounds the failure probability after i samples by $2^{-i/4}$, so $O(n)$ samples suffice.

Headline reproducible claim. For hidden subgroups $U \leq W_n$, Algorithm 7.1 recovers U with success probability $\geq 1 - 2^{-i/4}$ after i Fourier samples; in particular with $i = 4n$ expected samples the algorithm succeeds with probability $\geq 1 - 2^{-n}$, so the total number of oracle queries is polynomial (indeed linear) in $\log |W_n|$.

2 Claims table

ID	Claim	Type
C1	W_n is non-abelian; $[W_n:N] = 2$; every U factors as $(U \cap N) \cdot (U \cap U^t)$ (Lem. 4.1).	structural
C2	DFT_{W_n} has entries $(-1)^{\mu(g,h)} / \sqrt{ G }$, hence is a real orthogonal matrix.	algebraic
C3	Fourier sampling of a coset g_0U yields uniform samples from $U^\perp$ or $(U^t)^\perp$.	algorithmic
C4	After i samples the group $\langle U^\perp, (U^t)^\perp \rangle$ is generated with probability $\geq 1 - 2^{-i/4}$ (Lem. 6.3).	prob. bound
C5	Algorithm 7.1 recovers U with probability $\geq 1 - 2^{-n}$ using $O(n)$ queries.	headline
C6	DFT_{W_n} is implementable by $O(n)$ elementary gates (Fig. 3).	complexity

3 Method

3.1 Environment

- Host: CherryRd (macOS, python3 /usr/local/bin/python3, numpy 2.4.3).
- External libs: only numpy, standard library. No LLM, no GPU, no external service. Zero cost.
- PDF fetched from <https://arxiv.org/pdf/quant-ph/9812070> on 2026-07-05 into paper.pdf (16 pages, 166 896 bytes).
- Text extraction: pdftotext -layout (extraction/marker.md and extraction/nougat.mmd are pdftotext fallbacks; see failure_analysis.md for the honest note that Marker and Nougat are not installed on this host).

3.2 Implementation

The core is report/evidence/hsp_wreath.py. It implements:

1. enumerate_group(n) and elt_index(g, n): an explicit basis for the group algebra $\mathbb{C}[W_n]$ (used real, since DFT_{W_n} is real).
2. multiply(g1, g2): the wreath multiplication above.

3. `subgroup_closure(gens, n)`: BFS closure to enumerate a subgroup from generators. Verified against the paper’s Lemma 4.1 factorization on random inputs.
4. `mu(g1, g2, n)`: the pairing μ of Def. 4.3, all three cases ($a = a' = 0$, $a \oplus a' = 1$, $a = a' = 1$).
5. `build_dft(n)`: the $2^{2n+1} \times 2^{2n+1}$ Fourier matrix. **Sanity:** $\|FF^\top - I\|_\infty = 1.11 \times 10^{-16}$ at $n = 2$ (orthogonal to machine precision).
6. `sample_once(...)`: draws a uniform coset g_0U (equivalent to measuring the second register), forms the normalized coset state, applies DFT_{W_n} , and samples an outcome from $|\phi|^2$. Returns a single group element z .
7. `orthogonal_complement_via_phi(samples, n)`: applies the bijection $\phi : W_n \rightarrow \mathbb{F}_2^{2n+1}$ from Sec. 4.2 of the paper, computes the $\mathbb{F}_2$ -nullspace of the sampled matrix, and lifts the nullspace back to W_n (case split on the a -bit, respecting ϕ ’s swap).
8. `run_algorithm(...)`: the full pipeline of Algorithm 7.1 — draws i samples for the non-abelian half, runs an abelian HSP simulation on N (character-orthogonality on $\mathbb{F}_2^{2n}$) for $U \cap N$, and finally combines via the Lemma 4.1 factorization.

3.3 Experiments

Main run (results.json). $n = 2$, 6 trials (three random $|U| = 2$, three random $|U| = 4$), 10 reps each, 32 samples/rep ($= 4 \cdot 4n$).

Stress sweep (stress_results.json, section stress_n2). $n = 2$, five fixed subgroups (two abelian $|U| = 2$, two with transversal $a=1$ generators, one $|U| = 8$ mixed), sample counts $\{2, 4, 6, 8, 12, 16, 24, 32\}$, 20 reps each. Compares empirical success to Lemma 6.3 bound $1 - 2^{-i/4}$.

Non-abelian focus (nonabelian_focus_n2). Five subgroups whose generators include $a=1$ elements; these exercise the wreath-specific machinery (they contain transversal elements, so $|U \cap N| < |U|$ and the abelian subroutine alone does not recover them).

Scaling (scaling_n3). $n = 3$, $|W_n| = 128$; six random subgroups (four $|U| = 2$, two $|U| = 4$), 10 reps each, $4n = 12$ samples/rep — the paper’s expected count.

4 Results vs. paper

4.1 Main run at $n = 2$ (32 samples/rep, 10 reps)

Hidden subgroup	$ U $	Successes / reps	$\hat{p}$
$\langle((1, 1), (1, 1), 1)\rangle$	2	10/10	1.00
$\langle((1, 1), (1, 1), 0)\rangle$	2	10/10	1.00
$\langle((0, 1), (0, 0), 0)\rangle$	2	10/10	1.00
$\langle((1, 1), (0, 0), 0), ((1, 1), (1, 1), 0)\rangle$	4	10/10	1.00
$\langle((1, 1), (1, 1), 0), ((0, 1), (0, 0), 0)\rangle$	4	10/10	1.00
$\langle((1, 1), (0, 1), 0), ((1, 0), (0, 1), 0)\rangle$	4	10/10	1.00

DFT orthogonality error 1.1×10^{-16} at machine precision.

4.2 Stress sweep at $n = 2$ vs. Lemma 6.3 bound

Samples i	Paper bound $1 - 2^{-i/4}$	$\hat{p} (U =2, a=0)$	$\hat{p} (U =2, a=1)$	$\hat{p} (U =8 \text{ mixed})$
2	0.293	0.00	0.00	0.25
4	0.500	0.02	0.05	0.55
6	0.646	0.43	0.58	0.90
8	0.750	0.78	0.95	1.00
12	0.875	0.98	0.98	1.00
16	0.938	0.98	1.00	1.00
24	0.984	1.00	1.00	1.00
32	0.996	1.00	1.00	1.00

(Values shown for $|U| = 2$ cases are averaged over the two seeds per class in `stress_results.json`.) Empirical success is consistent with, and almost always exceeds, the paper’s worst-case Lemma 6.3 bound. The 0.9 threshold is crossed by ~ 8 samples, matching the paper’s $4n = 8$ prediction.

4.3 Non-abelian focus (transversal generators, $n = 2$)

All five subgroups (including the pathological $\langle(0, 0; 1)\rangle$) recovered at $\hat{p} = 1.00$ using 32 samples/rep, 20 reps. This confirms the wreath-specific machinery (Sec. 4.2–6 of the paper) works as claimed — the abelian-only subroutine cannot recover these subgroups on its own since $|U \cap N| < |U|$.

4.4 Scaling to $n = 3$ ($|W_n| = 128$)

Six random subgroups, 10 reps each, $4n = 12$ samples/rep: $\hat{p} = 1.00$ on every trial. The whole $n = 3$ pass finished in ~ 2.8 seconds wall-clock, confirming both correctness and that runtime is dominated by the small classical postprocessing rather than the group-algebra dimension.

5 Verdict

Verdict: REPLICATED.

Justification:

- The paper’s headline claim — Algorithm 7.1 recovers H with high probability using $O(n)$ quantum queries — is reproduced on both $n = 2$ and $n = 3$ instances, with empirical success $\hat{p} = 1.00$ using the paper’s own recommended sample budget $4n$ (and higher).
- The empirical success vs. sample-count curve tracks the paper’s Lemma 6.3 upper bound $1 - 2^{-i/4}$ on failure probability, with the 0.9 crossover at $i \approx 8$ (matching $4n$ at $n = 2$).
- Structural claims (Lemma 4.1 factorization, orthogonality of DFT_{W_n} , non-abelian Fourier sampling of cosets) all verified as computational sanity checks.
- Transversal-generator subgroups, which specifically exercise the non-abelian machinery, are recovered at the same rate as abelian ones. The claim genuinely covers non-abelian H , not just abelian corner cases.

What was *not* reproduced:

- The explicit circuit synthesis of DFT_{W_n} (Fig. 3) in $O(n)$ Toffoli gates. We used the $2^{2n+1} \times 2^{2n+1}$ matrix directly, which is faithful to the algorithm’s input/output behavior but does not verify the gate-count claim. This is a well-understood consequence of the recursive FFT construction of [?, ?] and is not the headline claim.
- Bounded-exponent and lattice-theoretic structural claims (Sec. 4) beyond what our closure routines exercise.

Open questions

Q1. Our empirical curve of success probability versus sample count i (Table above) is consistently *tighter* than the paper’s Lemma 6.3 bound $1 - 2^{-i/4}$, sometimes by 20+ percentage points at small i . Is there a sharper tail bound — possibly matching the abelian HSP’s $1 - 2^{-i}$ rate — for subgroup classes that are already balanced (Sec. 4.3), and if so, does the improvement extend to general subgroups after one round of “ t -completion”?

Q2. The paper’s Algorithm 7.1 uses one call to the abelian HSP subroutine on N per iteration *and* one non-abelian Fourier sample. Our implementation collapses this into a single sampling loop that provides evidence for both halves simultaneously. What is the minimum total number of oracle calls (as a function of n) if we allow this classical merging, and can it beat the paper’s $O(n) + O(n) = O(n)$ by a constant factor?

Q3. How badly does the algorithm degrade under a noisy oracle — e.g., a promise-violating f that agrees with a coset function on only a $(1 - \varepsilon)$ fraction of G ? The algorithm is exact in the promise model; our simulation reproduced $\hat{p} = 1.00$, but we did not perturb the oracle. Does the recovery probability decay smoothly in ε , or is there a sharp threshold at some $\varepsilon^*(n)$?

Q4. The paper hints (Sec. 8, "Conclusion and Outlook") that the technique may generalize to arbitrary split extensions $\mathbb{Z}_2^n \rtimes_{\varphi} \mathbb{Z}_2$. Which φ preserve the property that μ -orthogonality captures H ? In particular, does the "half of $U^{\perp}$ is in N " property that underlies Lemma 6.3 survive for non-swap actions, or do we lose the factor of $1/4$ in the exponent?

Q5. Our simulation recovered H at $\hat{p} = 1.00$ even for the pathological subgroup $\langle(0, 0; 1)\rangle$, whose generator is the central involution t . Are there “adversarial” subgroups $H \leq W_n$ (as n grows) for which the effective sample dimension of $\langle U^{\perp}, (U^t)^{\perp} \rangle$ is unusually small, causing the constant hidden in the $O(n)$ query bound to blow up? A worst-case scan across all balanced subgroups of W_3 (there are on the order of a few thousand) would settle this.

References

- [1] M. Rötteler and Th. Beth. Polynomial-time solution to the hidden subgroup problem for a class of non-abelian groups. *arXiv:quant-ph/9812070*, 1998.
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- [3] M. Püschel, M. Rötteler, Th. Beth. Fast quantum Fourier transforms for a class of non-abelian groups. Reference [12] in the paper.